❖ TOPIC:- HARDWARE

Module_2-Day_2:

❖ Tasks:

1. **List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.**

⇨

❖ SMPS (Switched Mode Power Supply):

Device	Why it uses SMPS
Desktop Computer	Converts AC power into the DC power required by internal components.
Laptop Charger	Converts AC electricity into suitable DC power to charge the laptop battery.
LED TV	Converts AC power into the DC voltages required by its electronic circuits.

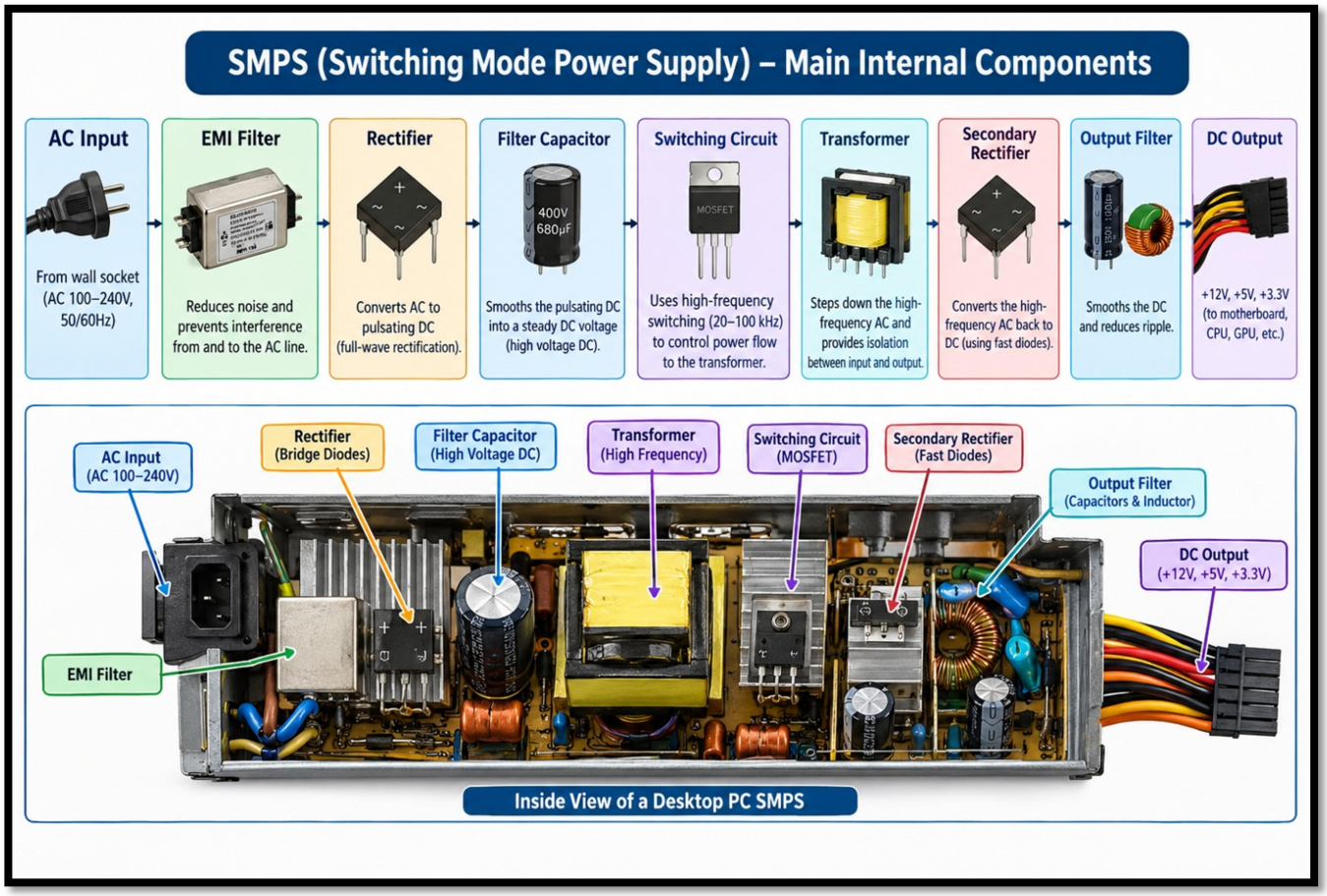
❖ UPS (Uninterruptible Power Supply):

Device	Why it uses UPS
Desktop Computer	Provides temporary backup power during a power cut and prevents sudden shutdowns.
Wi-Fi Router	Keeps the internet connection running during short power outages.

2. **Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.**

⇨



(1) AC Input: Receives AC power from the wall socket.
(2) EMI Filter: Reduces electrical noise and interference.
(3) Rectifier: Converts AC into pulsating DC.
(4) Filter Capacitor: Smooths the DC voltage.
(5) Switching Circuit: Switches power at high frequency.
(6) Transformer: Changes voltage and provides electrical isolation.
(7) Secondary Rectifier and Output Filter: Converts and smooths the high-frequency power into stable DC.
(8) DC Output: Supplies +12V, +5V, and +3.3V to computer components.

3. **Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.**

⇨

Common Power Connectors in a PC



ATX 24-pin
(Motherboard Connector)

Name	Devices It Powers	Safety Tip
ATX 24-pin (Motherboard Connector)	• Motherboard • CPU (through the motherboard power system)	Make sure the connector is fully seated and locked (clip snapped) before powering on. Do not force it, and always hold the connector (not the wires) when unplugging.



SATA Power Connector

Name	Devices It Powers	Safety Tip
SATA Power Connector	• HDD (Hard Disk Drive) • SSD (SATA) • Optical Drives • Some fans / accessories	Do not use excessive force. The connector only fits one way — align the L-shape and push gently until it is fully connected.



Molex Connector
(4-pin)

Name	Devices It Powers	Safety Tip
Molex Connector (4-pin)	• Older HDDs • Optical drives • Case fans • Some legacy devices	Make sure the connector is aligned properly (arrow/guide line on both sides). Do not force it, and avoid pulling on the wires.



PCIe 6/8-pin
(Graphics Card Connector)

Name	Devices It Powers	Safety Tip
PCIe 6/8-pin (Graphics Card Connector)	• Graphics Cards (GPU) • High-performance devices (e.g., some CPUs, PCIe cards)	Ensure the connector is fully locked and properly seated. Do not bend the pins and avoid using damaged cables.

Connector Name	Devices It Powers	One Safety Tip
ATX 24-pin	Motherboard	Ensure the locking clip is securely attached.
SATA Power	HDD, SATA SSD, optical drive	Align the L-shaped connector and never force it.
Molex 4-pin	Older HDDs, optical drives, some fans	Hold the connector when unplugging; never pull the wires.
PCIe 6/8-pin	Dedicated graphics card (GPU)	Ensure the connector is fully inserted and locked.

4. **Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.**

⇨

❖ Steps to Connect SMPS in a Gaming PC:

(1) **Turn off the power:** Switch off the SMPS, unplug the power cable, and ensure all components are safely installed.
(2) **Connect the motherboard:** Connect the 24-pin ATX power connector from the SMPS to the motherboard's main power socket.
(3) **Connect the CPU:** Connect the 4-pin or 8-pin EPS/CPU power connector to the CPU power socket near the processor.
(4) **Connect the graphics card (GPU):** Connect the required PCIe 6-pin/8-pin power cable or 12V-2x6 connector, depending on the graphics card model.
(5) Connect the storage devices:

- ★ **HDD/2.5-inch SATA SSD:** Connect a SATA power cable from the SMPS to the drive.
- ★ **M.2 NVMe SSD:** Usually, no separate power cable is needed because it receives power directly from the motherboard.

(6) **Connect other components:** Connect SATA power cables to compatible accessories such as fan hubs or RGB controllers, if required.
(7) **Check all connections:** Ensure every connector is fully seated and that no cables obstruct the fans.
(8) **Power on the PC:** Connect the AC power cable, switch on the SMPS, and start the PC. Check that the system boots correctly.